

Mathematics for Machine Learning

MAD-B3-2526-S2-MAT0611

Regularization via Bayesian Priors

Connecting to Last Session

Last time: Bayesian inference framework

- Prior beliefs $P(\theta)$
- Likelihood $P(D|\theta)$
- Posterior $P(\theta|D) \propto P(D|\theta) \cdot P(\theta)$

Today: Apply Bayesian thinking to **linear regression**

- What if we have **prior beliefs** about coefficients?
- Different priors $\rightarrow$ different regularization methods
- Bridge between Bayesian inference and optimization

Flow: Bayesian priors $\rightarrow$ MAP estimation $\rightarrow$ Ridge/LASSO/Elastic Net

Agenda

1. Bayesian Linear Regression & MAP Estimation

2. Gaussian Prior → Ridge (L2) Regularization

- Derivation from Bayesian principles
- Matrix perspective & properties

3. Laplace Prior → LASSO (L1) Regularization

- Why sparsity? Comparison with Gaussian
- Derivation & properties

4. Mixture Prior → Elastic Net

- Combining both priors for best of both worlds

5. Decision Framework: When to Use Each Method

Bayesian Linear Regression: The Setup

Standard linear model:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim N(0, \sigma^2 \mathbf{I})$$

Likelihood (same as before):

$$P(\mathbf{Y}|\boldsymbol{\beta}, \sigma^2) = N(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$$

NEW: Add prior beliefs about $\boldsymbol{\beta}$

$$P(\boldsymbol{\beta}) = \text{some distribution}$$

Posterior (Bayes' rule):

$$P(\boldsymbol{\beta}|\mathbf{Y}) \propto \underbrace{P(\mathbf{Y}|\boldsymbol{\beta})}_{\text{Likelihood}} \cdot \underbrace{P(\boldsymbol{\beta})}_{\text{Prior}}$$

What Does Bayesian Linear Regression Do?

Three key benefits:

1. Incorporates prior knowledge

- E.g., "coefficients should be small" → prevents overfitting
- E.g., "most coefficients are zero" → automatic feature selection

2. Provides full posterior distribution

- Not just point estimate $\hat{\beta}$
- Get uncertainty: $P(\beta | \mathbf{Y})$

3. Natural regularization

- Prior acts as "soft constraint"
- Controls model complexity

Maximum A Posteriori (MAP) Estimation

Computing the entire posterior distribution $P(\boldsymbol{\beta}|\mathbf{Y})$ is often complex

MAP simplification: Finding the mode (most probable value) of the posterior

$$\hat{\boldsymbol{\beta}}_{\text{MAP}} = \arg \max_{\boldsymbol{\beta}} P(\boldsymbol{\beta}|\mathbf{Y})$$

Taking logs (easier to optimize):

$$\hat{\boldsymbol{\beta}}_{\text{MAP}} = \arg \max_{\boldsymbol{\beta}} \{ \log P(\mathbf{Y}|\boldsymbol{\beta}) + \log P(\boldsymbol{\beta}) \}$$

Or minimize negative log:

$$\hat{\boldsymbol{\beta}}_{\text{MAP}} = \arg \min_{\boldsymbol{\beta}} \left\{ \underbrace{-\log P(\mathbf{Y}|\boldsymbol{\beta})}_{\text{Loss term}} + \underbrace{(-\log P(\boldsymbol{\beta}))}_{\text{Regularization}} \right\}$$

Gaussian Prior on Coefficients

Why Gaussian prior? To keep coefficients small and avoid overfitting

Prior specification:

$$\beta_j \sim N(0, \tau^2) \quad \text{independently for each } j$$

or matricially:

$$\boldsymbol{\beta} \sim N(\mathbf{0}, \tau^2 \mathbf{I})$$

where $\boldsymbol{\beta}$ is the vector of coefficients and $N(\mathbf{0}, \tau^2 \mathbf{I})$ is a multivariate normal distribution with mean vector $\mathbf{0}$ and covariance matrix $\tau^2 \mathbf{I}$.

Gaussian Prior on Coefficients

Properties:

- Most probability mass near zero
- Favours small coefficients
- Allows non-zero values (soft constraint)
- Variance τ^2 controls strength: small $\tau \rightarrow$ strong prior

When to use:

- Believe all features somewhat relevant
- Want to reduce overfitting
- Have many correlated predictors

Deriving the MAP with Gaussian Prior

Start with Gaussian prior:

$$P(\boldsymbol{\beta}) = \prod_{j=1}^p N(\beta_j | 0, \tau^2) \propto \exp \left(-\frac{1}{2\tau^2} \sum_{j=1}^p \beta_j^2 \right)$$

Gaussian likelihood (assuming known σ^2):

$$P(\mathbf{Y} | \boldsymbol{\beta}) \propto \exp \left(-\frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 \right)$$

MAP estimation (negative log):

$$\hat{\boldsymbol{\beta}}_{\text{MAP}} = \arg \min_{\boldsymbol{\beta}} \left\{ \frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \frac{1}{2\tau^2} \sum_{j=1}^p \beta_j^2 \right\}$$

Ridge regression

$$\hat{\boldsymbol{\beta}}_{\text{MAP}} = \arg \min_{\boldsymbol{\beta}} \left\{ \frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \frac{1}{2\tau^2} \sum_{j=1}^p \beta_j^2 \right\}$$

Setting $\lambda = \sigma^2 / \tau^2$ (scaling constants):

$$\hat{\boldsymbol{\beta}}_{\text{Ridge}} = \arg \min_{\boldsymbol{\beta}} \left\{ \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\}$$

A Gaussian prior is equivalent to an L2 penalty to a linear regression, also known as **Ridge regression**

Ridge Regression: Matrix Perspective

Ridge objective:

$$\hat{\boldsymbol{\beta}}_{\text{Ridge}} = \arg \min_{\boldsymbol{\beta}} \{ \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \}$$

Closed-form solution (take derivative, set to zero):

$$\hat{\boldsymbol{\beta}}_{\text{Ridge}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{Y}$$

$\lambda \mathbf{I}$ "regularizes" the covariance matrix. If $\lambda = 0$, then we get the classic OLS estimator:

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

Ridge Regression: Properties

Ridge regression is useful to reduce overfitting and handle multicollinearity, which typically causes instability in OLS estimates and very large coefficients.

- Always invertible: $\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I}$ is positive definite even if $\mathbf{X}^T \mathbf{X}$ is singular
- Shrinks all coefficients toward zero, but doesn't set any exactly to zero
- Control via λ :
 - $\lambda = 0 \rightarrow$ OLS (no regularization)
 - $\lambda \rightarrow \infty \rightarrow \beta \rightarrow \mathbf{0}$

If we want a regularization technique that sets some coefficients exactly to zero (so we can remove the associated variables from the model), we need a different prior.

Laplace Prior on Coefficients

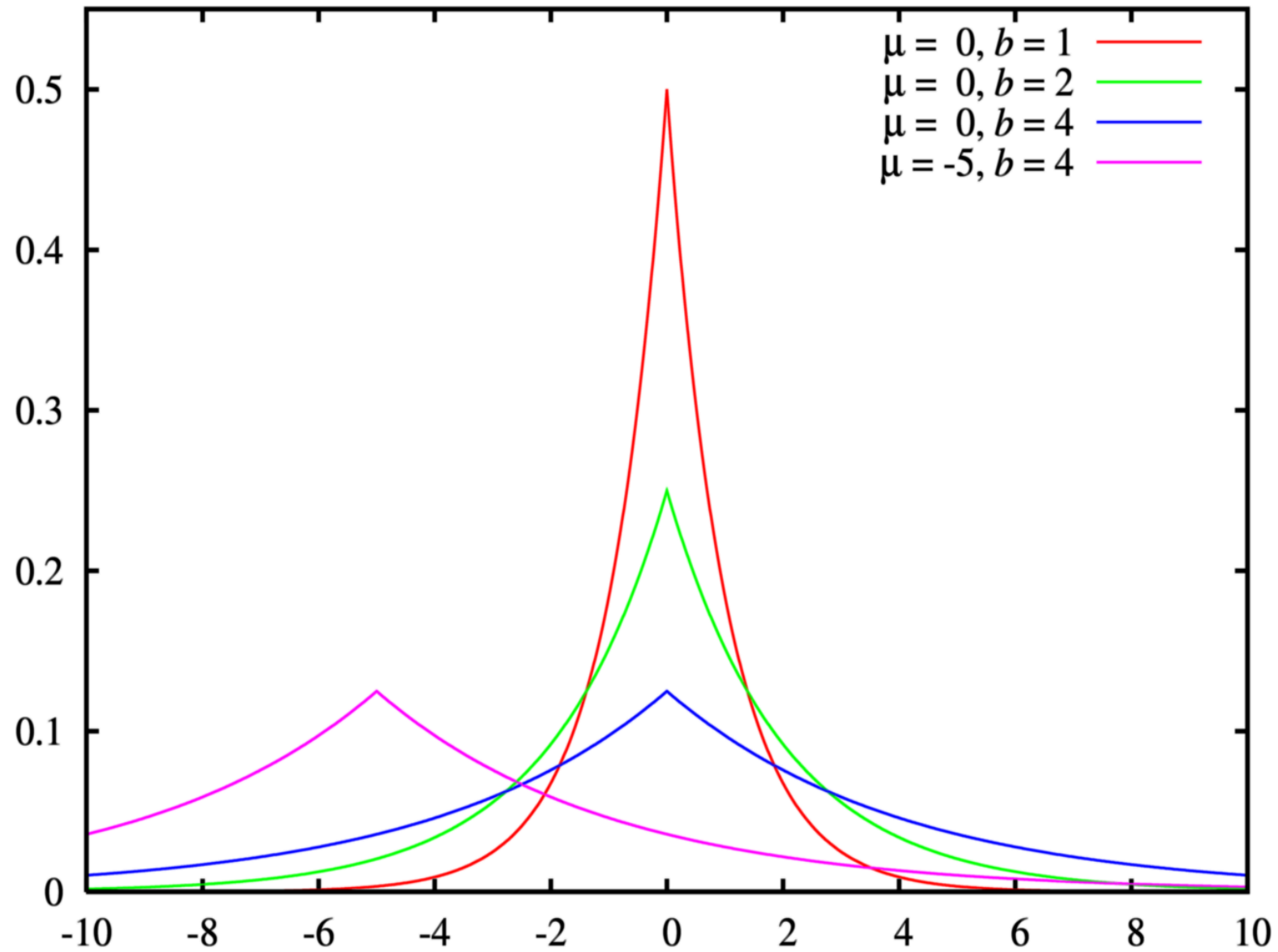
Why Laplace prior? To encourage sparsity and perform feature selection

Prior specification:

$$P(\beta_j) = \frac{1}{2b} \exp \left(-\frac{|\beta_j|}{b} \right) \quad \text{independently}$$

Properties:

- **Sharp peak at zero** → strong preference for exactly zero
- Heavy tails → allows some large values
- Parameter b controls scale



Comparison with Gaussian

Prior	Shape	Effect
Gaussian	Smooth bell	Continuous shrinkage
Laplace	Sharp peak	Sparsity

When to use:

- Believe many coefficients are exactly zero
- Want automatic feature selection
- Need interpretable model (few features)

Deriving MAP with Laplace Prior

Start with Laplace prior:

$$P(\boldsymbol{\beta}) = \prod_{j=1}^p \frac{1}{2b} \exp\left(-\frac{|\beta_j|}{b}\right) \propto \exp\left(-\frac{1}{b} \sum_{j=1}^p |\beta_j|\right)$$

Gaussian likelihood (assuming known σ^2):

$$P(\mathbf{Y}|\boldsymbol{\beta}) \propto \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2\right)$$

MAP estimation (negative log):

$$\hat{\boldsymbol{\beta}}_{\text{MAP}} = \arg \min_{\boldsymbol{\beta}} \left\{ \frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \frac{1}{b} \sum_{j=1}^p |\beta_j| \right\}$$

LASSO Regression

$$\hat{\boldsymbol{\beta}}_{\text{MAP}} = \arg \min_{\boldsymbol{\beta}} \left\{ \frac{1}{2\sigma^2} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \frac{1}{b} \sum_{j=1}^p |\beta_j| \right\}$$

Setting $\lambda = 2\sigma^2/b$:

$$\hat{\boldsymbol{\beta}}_{\text{LASSO}} = \arg \min_{\boldsymbol{\beta}} \left\{ \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

A Laplace prior is equivalent to an L1 penalty to a linear regression, also known as LASSO (Least Absolute Shrinkage and Selection Operator) regression

LASSO Regression: Properties

Objective:

$$\hat{\boldsymbol{\beta}}_{\text{LASSO}} = \arg \min_{\boldsymbol{\beta}} \{ \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|_1 \}$$

Properties:

- No closed-form solution (requires iterative optimization methods)
- Produces sparse models
 - Sets some coefficients **exactly to zero** → automatic feature selection
- As λ increases, more coefficients become zero
 - Least important variables get zeroed out first as λ increases

Why not both?: Laplace + Gaussian combined prior

- Ridge: Handles correlated features well, stable
- LASSO: Sparse solutions, feature selection

Combined prior (unnormalized):

$$P(\beta_j) \propto \underbrace{\exp\left(-\frac{\beta_j^2}{2\tau^2}\right)}_{\text{Gaussian}} \cdot \underbrace{\exp\left(-\frac{|\beta_j|}{b}\right)}_{\text{Laplace}}$$

Taking negative log like before:

$$-\log P(\boldsymbol{\beta}) \propto \frac{1}{2\tau^2} \|\boldsymbol{\beta}\|_2^2 + \frac{1}{b} \|\boldsymbol{\beta}\|_1$$

This naturally leads to Elastic Net!

Elastic Net Regression

Objective (combines L1 and L2):

$$\hat{\boldsymbol{\beta}}_{\text{ElasticNet}} = \arg \min_{\boldsymbol{\beta}} \left\{ \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda_1 \|\boldsymbol{\beta}\|_1 + \lambda_2 \|\boldsymbol{\beta}\|_2^2 \right\}$$

Alternative parameterization (sklearn uses this):

$$\min_{\boldsymbol{\beta}} \left\{ \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \left(\alpha \|\boldsymbol{\beta}\|_1 + \frac{1 - \alpha}{2} \|\boldsymbol{\beta}\|_2^2 \right) \right\}$$

- $\alpha \in [0, 1]$: mixing parameter ($\alpha = 1 \rightarrow \text{LASSO}$, $\alpha = 0 \rightarrow \text{Ridge}$)
- $\lambda > 0$: overall regularization strength

Best of both worlds:

- Sparsity from L1 (feature selection)
- Stability from L2 (grouped selection for correlated features)

When to Use Each Method?

Method	Prior	Use When	Benefits
Ridge	Gaussian	All features relevant, many correlated	Stable, closed-form
LASSO	Laplace	Many irrelevant features	Sparse, interpretable
Elastic Net	Mixture	High correlation + need sparsity	Balanced

Decision tree:

1. Need sparsity/interpretability? → **LASSO or Elastic Net**
2. $p > n$ (more features than samples)? → **Ridge or Elastic Net**
3. Features highly correlated? → **Ridge or Elastic Net**
4. Few features matter? → **LASSO**
5. Uncertain? → **Elastic Net** (robust choice)

Summary: Bayesian perspective unifies regularization

1. Prior choice determines regularization type

- Gaussian prior → Ridge (L2 penalty)
- Laplace prior → LASSO (L1 penalty)
- Combined prior → Elastic Net (L1 + L2)

2. MAP estimation = penalized optimization

- Stronger prior (smaller variance) = more regularization (higher λ)

3. Each method serves different goals

- Ridge: stability, handles correlation
- LASSO: sparsity, feature selection
- Elastic Net: balanced approach

References & Further Reading

Core Textbooks:

- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning* (2nd ed.). Springer.
 - Chapter 3.4: Shrinkage Methods
- Murphy, K. P. (2012). *Machine Learning: A Probabilistic Perspective*. MIT Press.
 - Chapter 7.5: Ridge Regression, Chapter 13.3: Sparse Linear Models
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning* (2nd ed.). Springer.
 - Chapter 6: Linear Model Selection and Regularization